

DILIPKUMAR NEHRU

Backend Engineer • Python & Node.js • Scalable Systems for AI/Analytics Platforms

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PROFESSIONAL SUMMARY

Backend Engineer with 4+ years designing and maintaining scalable, high-performance server-side systems for enterprise CRM and AI analytics platforms, using Node.js, asynchronous programming, and microservices architecture. Experienced across the full backend lifecycle: API design, database performance optimisation, message-queue based async processing (RabbitMQ), and production support for systems handling 50,000+ daily requests. Comfortable working full-stack with React/Next.js, collaborating closely with data and product teams, and applying Python for scripting and tooling. Track record of code-quality ownership (code reviews, GitFlow, clean-code standards) and mentoring through QA/dev handoffs, with a strong debugging and problem-solving foundation from a prior QA automation background.

KEY HIGHLIGHTS RELEVANT TO BACKEND / AI LABS ROLE

- ▶ Designed and maintained scalable RESTful APIs and event-driven microservices (Node.js, Express.js, NestJS, RabbitMQ) serving 50,000+ daily requests at 99.5% uptime.
- ▶ Collaborated cross-functionally with data, frontend, and product teams in 2-week Agile sprints to ship features supporting analytics and CRM platforms — directly analogous to supporting model development and productization with Data Scientists/ML Engineers.
- ▶ Optimised backend performance and resolved bottlenecks: cut MongoDB query latency by 35% and async job completion time from 8s to under 2s through queue redesign and indexing.
- ▶ Full-stack exposure across React.js/Next.js and Tailwind CSS, enabling end-to-end feature delivery alongside frontend engineers.
- ▶ Enforced code quality through structured code reviews, GitFlow branching, and clean-code standards, reducing post-merge defects by 25%.
- ▶ Working knowledge of Python for scripting/automation; strong foundation in asynchronous, reliable backend design patterns transferable to Python and Golang service development.

WORK EXPERIENCE

Backend Engineer – Software Developer — Triotech Bizserve Private Limited

Jun 2026 – Sep 2026

- ▶ Developed and enhanced backend services for Ringingo, a multi-tenant cloud telephony SaaS platform supporting real-time CDR processing, webhook delivery, and API integrations.
- ▶ Designed an asynchronous, event-driven CDR pipeline (RabbitMQ) with timezone-aware scheduled webhooks, incremental cursor-based delivery, and retry/delivery-log mechanisms — reusable patterns for reliable, scalable backend architecture.
- ▶ Built token-to-API-key management with JWT-based authentication (x-api-key and Bearer mechanisms), including key refresh and expiry handling.
- ▶ Delivered end-to-end features across NestJS/Node.js, MongoDB, microservices, and React/Next.js with Tailwind CSS for secure, multi-tenant applications.

Software Engineer – Backend Developer — Zuqo Corporation India Pvt Ltd

Dec 2023 – May 2025

- ▶ Designed and maintained scalable RESTful APIs using Node.js and Express.js, serving 50,000+ daily requests for enterprise web applications with 99.5% uptime.
- ▶ Reduced MongoDB query response times by 35% through strategic indexing, query optimisation, and connection pooling on the NovoNordisk CRM platform.
- ▶ Implemented asynchronous background task processing with RabbitMQ, cutting average job completion time from 8s to under 2s for UrbanPiper AI Analytics.
- ▶ Integrated MongoDB and MySQL with backend microservices, standardising schema design patterns and reducing inter-service data inconsistencies by 60%.
- ▶ Collaborated with a 6-member cross-functional team in 2-week Agile sprints, consistently delivering features on schedule with zero critical production rollbacks.
- ▶ Debugged and resolved 30+ production incidents, improving stability and reducing MTTR by ~45%.
- ▶ Enforced clean code standards, GitFlow branching, and code review practices, reducing post-merge defect rate by 25%.

- ▶ Implemented JWT authentication and OAuth 2.0 authorisation flows across multiple APIs, eliminating session vulnerabilities and enabling secure third-party integrations.
- ▶ Integrated Socket.io for real-time WebSocket notifications and live dashboard updates, reducing client-side polling overhead by 80%.

Software Test Engineer – QA Automation — Zuqo Corporation India Pvt Ltd

Apr 2022 – Dec 2023

- ▶ Built and maintained a Selenium automation suite covering 200+ functional and regression test cases, cutting manual testing effort by 40%.
- ▶ Performed API testing and backend validation using Postman and JMeter, identifying 80+ defects before production deployment.
- ▶ Created structured bug reports, test plans, and troubleshooting runbooks adopted as the QA documentation standard across the team.
- ▶ Collaborated directly with backend developers to triage and resolve defects, reducing average bug resolution cycle from 5 days to 2 days.

PROJECT HIGHLIGHTS

NovoNordisk CRM — Healthcare Enterprise Platform

- ▶ Architected scalable backend APIs (Node.js, Express.js, MongoDB) handling 50,000+ daily requests for a global pharmaceutical CRM.
- ▶ Implemented JWT authentication, OAuth 2.0, field-level encryption, and RBAC for HIPAA-aligned secure data handling.
- ▶ Optimised 15 high-frequency MongoDB aggregation queries, reducing dashboard load times by 35% for 500+ concurrent users.

UrbanPiper AI Analytics — F&B Analytics Dashboard

- ▶ Built event-driven backend microservices (RabbitMQ, Socket.io) to process and push real-time analytics data streams to 100+ restaurant dashboards.
- ▶ Reduced async job processing latency from 8s to under 2s by redesigning queue consumers and adding dead-letter queue error handling.
- ▶ Delivered REST APIs consumed by React analytics dashboards, enabling real-time KPI monitoring with sub-500ms response times.

TECHNICAL SKILLS

Languages	JavaScript (ES6+), Node.js, TypeScript (basic), Python, SQL, Java
Frameworks	NestJS, Express.js, React.js, Next.js, Socket.io
Databases	MongoDB, MySQL, PostgreSQL
Async / Microservices	RabbitMQ, Event-Driven Architecture, WebSockets, CDR Processing, Design Patterns
Auth & Security	JWT, OAuth 2.0, bcrypt, HTTPS/TLS
Backend	RESTful APIs, Microservices, API Integration, Async Programming, Webhooks, Retry Mechanisms
Frontend	React.js, Next.js, Tailwind CSS
Testing	Postman, Selenium WebDriver, JMeter, API Automation
DevOps / Tools	Git, Bitbucket, GitHub, Linux CLI, CI/CD awareness
Practices	Agile Methodology, Code Reviews, Mentorship, Cross-functional Collaboration

CERTIFICATIONS

- ▶ Selenium Automation & Manual Testing — QSpiders, Bangalore (2022)
- ▶ MongoDB Basics (M001) — MongoDB University (Recommended Next)
- ▶ Node.js Application Developer — OpenJS Foundation (Recommended Next)

EDUCATION

Bachelor of Engineering – Electronics & Communication Engineering | RVS Technical Campus, Coimbatore (Anna University) | 2020

Diploma – Electronics & Communication Engineering | Krishnasamy Memorial Polytechnic College, Cuddalore | 2012–2015